

Control Flow Practice Question Set

(NOTE: PLEASE TAKE ALL THE INPUT FROM USER.)

1. If statement

1. Write a program that checks if a number is positive. If it is, print "The number is positive".
2. Write a program to check if a person is eligible to vote. The person must be **18 or older** and a **citizen of the country**. Use `age` and `isCitizen` boolean flag as input. Print "Eligible to vote" or "Not eligible to vote".
3. Write a program that checks if a day is a **weekend**. Input: `day` (string). If the day is "Saturday" or "Sunday", print "Weekend".

2. If else Statement

1. Write a program to check if a number is even or odd. Use `if-else` to print "Even" or "Odd".
2. Write a program to determine if a person is eligible to vote.
Input: Age of the person
Output: "Eligible to vote" if age is 18 or above, otherwise "Not eligible".
3. Write a program to compare two numbers and print which one is greater, or if they are equal.
Input: Two integers
Output: "First is greater", "Second is greater", or "Both are equal"
4. Write a program to find the largest among three given numbers using `if-else`.
Input: Three integers
Output: "First is largest", "Second is largest", or "Third is largest".
5. Write a program to assign grades based on marks using the following rules:
 - 90–100 → "Grade A"
 - 80–89 → "Grade B"
 - 70–79 → "Grade C"
 - 60–69 → "Grade D"
 - Below 60 → "Grade F"**Input:** Marks as an integer
Output: Grade as per the range
6. Write a program to simulate a login system.
Input: A username and a password
Output: If the username is "admin" and the password is "1234", print "Login successful"; otherwise, print "Invalid credentials"
7. Write a program to check if a given year is a leap year or not. **Conditions:**

- A year is a leap year if it is divisible by 4 and not divisible by 100, or divisible by 400.

Input: Year as an integer

Output: "Leap Year" or "Not a Leap Year"

8.

Write a program to check whether three given sides can form a valid triangle.

Condition: The sum of any two sides must be greater than the third.

Input: Three integers representing sides

Output: "Valid triangle" or "Invalid triangle"

9. Write a program to calculate movie ticket price after applying discounts.

Input:

- Age of the customer
- Day of the week

Discount Rules:

- If age < 12 or age > 60 → apply 50% discount
- If day is "Wednesday" → apply 30% discount
- Discounts don't combine — choose the best one

Output: Final ticket price after applying the appropriate discount

10. Write a program to decide whether a vehicle should go or stop based on signal color and vehicle type.

Input:

- Signal color ("red", "yellow", "green")
- Vehicle type ("ambulance" or "normal")

Rules:

- If signal is "red" and vehicle is "normal" → Stop
- If signal is "yellow" → Slow down
- If vehicle is "ambulance" → Go regardless of signal
- Else → Go

Output: Appropriate action: "Stop", "Go", or "Slow down"

3. If else if statements

1. Write a program to assign a grade based on marks using the following conditions:

- 90–100 → "A"
- 80–89 → "B"

- 70–79 → "C"
- 60–69 → "D"
- Below 60 → "F"

Input: Marks (0 to 100)

Output: Grade

2. Write a program that takes a number from 1 to 7 and prints the corresponding day of the week.

Example: 1 → "Monday", 7 → "Sunday"

Input: Integer (1–7)

Output: Name of the day

3.

Write a program to display a message based on temperature:

- Temp < 0 → "Freezing cold"
- 0–10 → "Very Cold"
- 11–20 → "Cold"
- 21–30 → "Pleasant"
- 31–40 → "Hot"
- 40 → "Very Hot"

Input: Temperature (integer)

Output: Message

4. Write a program to calculate electricity bill based on units consumed:

- ≤ 100 units → ₹1 per unit
- 101–200 units → ₹2 per unit
- 201–300 units → ₹3 per unit
- 300 units → ₹5

per unit **Input:**
Units consumed

Output: Total bill

5. Write a program that takes BMI as input and classifies the person:

- < 18.5 → "Underweight"
- 18.5–24.9 → "Normal"
- 25–29.9 → "Overweight"
- 30–34.9 → "Obese Class I"
- ≥ 35 → "Obese

Class II" **Input:**

BMI (double)

Output: Category

6. Write a program that takes a student's percentage and prints:

- ≥ 90 → "Excellent"
- 80–89 → "Very Good"
- 70–79 → "Good"
- 60–69 → "Average"
- < 60 → "Needs Improvement"

Input: Percentage

Output: Performance level

7. Write a program to calculate income tax based on the following slabs:

- $\leq ₹2,50,000 \rightarrow$ "No tax"
- $₹2,50,001 - ₹5,00,000 \rightarrow$ "5% tax"
- $₹5,00,001 - ₹10,00,000 \rightarrow$ "20% tax"
- $₹10,00,000 \rightarrow$ "30%

tax" **Input:** Annual

income **Output:**

Applicable tax rate

8. Write a program to check if a student passed based on marks in three subjects. Rules:

- If all subjects ≥ 40 and average $\geq 50 \rightarrow$ "Pass"
- Else if any one subject $< 40 \rightarrow$ "Fail in that subject"
- Else $\rightarrow$ "Fail due to low

average" **Input:** Marks of 3

subjects **Output:** Result

9.

Write a program that takes the vehicle speed as input and prints:

- $\leq 40 \rightarrow$ "Safe speed"
- $41-60 \rightarrow$ "Drive carefully"
- $61-80 \rightarrow$ "Over-speeding! Warning"
- $80 \rightarrow$ "Challan will be issued"

Input: Speed in km/h

Output: Message based on speed

10. Write a program to calculate water charges:

- ≤ 30 liters $\rightarrow$ "₹10"
- $31-100$ liters $\rightarrow$ "₹20"
- $101-200$ liters $\rightarrow$ "₹40"
- 200 liters $\rightarrow$ "₹70"

Input: Water consumption in liters

Output: Water bill amount

4. Ternary Operator:

1. Write a program to check if a number is even or odd using a ternary operator.

- Input: An integer
- Output: "Even" or "Odd"

2. Write a program to find the maximum of two numbers using a ternary operator.

- Input: Two integers
- Output: Larger number

3. Write a program to check whether a number is positive or negative (or zero).
 - Input: Integer
 - Output: "Positive", "Negative", or "Zero"
4. Write a program to check if a year is a leap year using ternary.
 - Input: Year
 - Output: "Leap year" or "Not a leap year"
5. Write a program to check if a student passed (marks ≥ 40) using ternary.
 - Input: Marks
 - Output: "Pass" or "Fail"
6. Write a program to find the minimum of three numbers using nested ternary.
 - Input: Three integers
 - Output: Minimum value
7. Write a program to find the maximum of three numbers using nested ternary.
 - Input: Three integers
 - Output: Minimum value
8. Write a program that checks if a user is allowed to log in:
 - If username is not empty and password is not empty $\rightarrow$ "Login Allowed"
 - Else $\rightarrow$ "Login Denied"
9. Write a program to check if a person can vote based on age and citizenship:
 - Age ≥ 18 and is a citizen $\rightarrow$ "Eligible"
 - Else $\rightarrow$ "Not Eligible"
10. Check scholarship status based on percentage and income:
 - If percentage ≥ 90 and income $< ₹2L \rightarrow$ "Full Scholarship"
 - Else if percentage ≥ 80 or income $< ₹4L \rightarrow$ "Partial Scholarship"
 - Else $\rightarrow$ "No Scholarship"

5. Switch Cases:

1. Write a program that takes a number (1–7) and prints the corresponding day of the week.
 - Input: Integer from 1 to 7
 - Output: "Monday", "Tuesday", ..., "Sunday"
2. Write a program that takes a character and checks if it's a vowel or consonant using `switch-case`.
 - Input: A single alphabet character
 - Output: "Vowel" or "Consonant"
3. Write a program to print remarks based on grade letter:
 - A $\rightarrow$ Excellent

- B → Good
 - C → Average
 - D → Poor
 - F → Fail
 - Any other → Invalid Grade
4. Write a basic calculator using **switch-case** that supports +, -, *, /.
- Input: Two numbers and an operator
 - Output: Result of the operation
5. Write a program that takes a number (1–12) and prints the name of the month.
- Input: Integer from 1 to 12
 - Output: "January", "February", ..., "December"
6. Write a program that displays an ATM menu with options: Check Balance, Withdraw, Deposit, Exit. Use **switch-case** to perform the selected operation.
- Input: Option (1–4)
 - Output: Corresponding operation
7. Write a program to print number of days in a month (use switch for 1–12):
- Input: Month number
 - Output: "31 Days", "30 Days", or "28/29 Days" for February
 - Optional: Ask user for year too, and check for leap year (use if inside switch)
8. WAP to Create a menu: Use **switch-case** to perform the selected calculation
1. Area of Circle
 2. Area of Rectangle
 3. Area of Triangle
 4. Exit
- Input: User's choice, and respective dimensions
 - Output: Area
- 9.
- Write a program that accepts a product code (1–5), quantity, and uses **switch-case** to calculate total price based on predefined product rates.
- Input: Product code and quantity
 - Output: Total bill

Use **switch** to assign price per unit based on product code.

10. Write a program to select a travel package:
- 1 → Goa (₹15,000)
 - 2 → Manali (₹10,000)
 - 3 → Kerala (₹12,000)
 - 4 → Exit

Calculate cost for **n** travelers and apply discount:

- If $n > 5 \rightarrow 10\%$ discount
- Use `switch-case` for destination and `if` for discount

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